

Named Entity Recognition in Malayalam using Fuzzy Support Vector Machine

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Abstract—Named entities in a text are the atomic elements that represent the name of something, and the name can be a person name, name of an organization, name of a place or location etc. In the field of information extraction the identification and classification of named entities are quite an important task. The identification and classification of the named entities in a text into some pre-defined classes is known as named entity recognition. The commonly used pre-defined classes are the person name, place, name of organization, date, numerals, measurements, and so on. It is a subtask of information extraction and many other natural language processing like text summarization, text categorization, question answering etc. In the case of language processing of Malayalam documents no effective tools are readily available. Through this paper a named entity recognition for Malayalam language is presented. The system proposed follows machine learning approach using support vector machine integrated with fuzzy module for improving the performance. The design is a kind of One-Against-All-Multi classification technique to solve the ambiguity caused by traditional SVM classifier. The system is based on contextual semantic rules and linguistic grammar rules. Malayalam NER is a challenging work as there is no specific feature for identifying named entities like capitalization feature in English. Also no named entity tagged corpus for Malayalam language is available for training the system. The system defines four primary named entity classes, i.e, Name, Organization, Place and Date.

Keywords—NER, NE class, SVM, Fuzzy.

I. INTRODUCTION

The single element in a text which may belong to a class of names like person name, organization name, place name, or may belong to a date class, numerical class, or expressions of time or measurement class. The task of identifying such named entities is known to be Named Entity Recognition or NE [1]. The various classes of the named entities are predefined as per the nature and requirement of the task or process. For example when the task deals with the numerical data, then the categories related to numerical data are defines and the NER handles the numerical data. Similarly, if geographical classification is important, it may be necessary to classify each location entity as a particular type of location [2]. Named Entity Recognition is sometimes also known as named entity identification [3] or named entity chunking or named entity extraction. The identification and classification of the words or phrases in

a document as named entities is the prior function of NER. Named Entity identification involves marking the named entity in an input text and classification included categorising the entity to a particular class according to its role in the context.

Many works in the field of natural language processing like named entity recognition, POS tagging, summarization, categorization etc have been done for English and European Languages. The approaches done so far may be supervised and unsupervised which are widely accepted in those languages. Works on NER in many Indian languages such as Tamil, Bengali, Hindi, Malayalam, Assamese etc. are on its way. Unlike English, NER in Malayalam is difficult as it lacks certain features like capitalization. Also the Indian languages are mostly morphologically rich, agglutinative and resource scarce language which makes the study on it difficult.

The major difficulties are that the atomic elements which are the named entities may firstly be difficult to locate, and once identified, difficult to classify [2]. And also making distinctions based on world knowledge is difficult (eg. organization name or person name)[4]. The system proposed in the paper works on Malayalam language, which is an Indian language. Indian languages belong to several major language families like the Indo-European languages, Indo-Aryan and the Dravidian languages. The difficulties in recognizing NEs occur due to some language properties [5]. Some among them are:

- Rich morphology - Root word identification from its compound word is a difficult task. Language has a rich morphological variations for a single root word.
- Lack of Capitalisation - The words that begins with a capital letter is probably a named entity. This feature is not available in any of the Indian languages.
- Semantic Uncertainty - Uncertainty between common and proper nouns resulting in ambiguity. Eg: common words such as Roja can be a flower or a person name.
- Freedom of Spells - It is commonly accepted to spell the named entities differently. For example : In Tamil person name -Roja is spelt as rosa, roja.
- Sentence structure - A specific sentence structure is absent for many Indian languages.
- Resource scarce - Most of the Indian languages have less resources. Unavailability of language processing tools is a major challenge in this field.
- Efforts in the field of named entity recognition in Indian

languages is very less.

This system presented in this paper is a supervised machine learning approach for identifying and classifying the named entities in a Malayalam text. For the purpose of identifying the named entities contextual semantic features and linguistic grammar features are used as rules. The system uses support vector machine tool for classification of named entities into predefined classes, here which are Name, Organization, Place and Date class. Support Vector Machine(SVM) is a supervised machine learning approach for classification of statistical data. It solves binary classification problem so called a linear classifier but can be extended to a multi classifier. The features are mapped to SVM feature vectors and binary values are given to the features by setting true(1) if it exist and false(0) otherwise. These feature vectors are the decision parameters that identify the class of the named entities. Identifying and classifying the named entities are not much a complicated task, but still it faces many challenges. The challenges arise mainly due to the agglutinative nature and the semantic ambiguity of the language. Organization and person names can be the same, and follow similar formatting [2]. To handle the situation a fuzzy module is integrated which produce a One-Against-All-Multi classification and hence improves the performance of the system.

The content of the paper is divided into three sections. In the next section some of the previous works in the named entity recognition are discussed. The works may include analyzing the challenges in the field and various approaches used in many systems and their performances. In the following sections the proposed system with the system design and the methodology of the system are explained. Final section includes the conclusion with the evaluation results.

II. RELATED WORKS

The authors Bindu.M.S and Sumam Mary Idicula discussed their work [3], on Named Entity (NE) Classifier created by using Multiclass Support Vector Machines based on linguistic grammar principles. The system uses inner word as internal feature and context word as external features are used. The multiclass SVM each named entity is classified into the most appropriate category of named entity. Another paper by Alireza Mansouri , Lilly Suriani Affendey and Ali Mamat employed Fuzzy Support Vector Machine algorithm [6], a machine learning approach for recognizing and classifying the named entities. In the paper the authors solved the problem of multi-classification of the NER systems by a fuzzy membership function thus removing the Support Vector Machines weakness points in NER precision and multi-classification. Thus the technique is a kind of One-against-All multi-classification technique [7].

Bindu M. S. and Sumam Mary Idicula proposed an NER system for Malayalam language using statistical approach in their paper [3]. The system uses a knowledge based approach in which a dictionary is used to check if each token is in the dictionary, if the token does not belong to the dictionary then uses a Tag Disambiguator. The system categorize an entity as either sole-entity, Constituent-entity, Dependant-entity or Not-an-entity. The NER system proposed by Jisha P Jayan ,Rajeev R R and Elizabeth Sherly is a Named Entity Recognition for Malayalam language using statistical

approach. The authors presented a properly implemented approach for the identification of atomic units of the text into named entities and for the classification task, a supervised machine learning method using TnT based on a Hidden Markov Model and Viterbi algorithm for Malayalam language [8].

Apurbalal Senapati, Arjun Das and Utpal Garain[9] describes two systems for Named Entity Recognition (NER) and the performance of the systems have been compared. The first system is a rule-based one whereas the second one is statistical (based on CRF) in nature. The authors concluded that the rule based approach shows better performance than the statistical model. N.Jeyashenbagavalli, Dr.K.G.Srinivasagan and S.Suganthi in their work implemented a named entity recognizer using hybrid approach[10]. Hybrid approach means a combination of two or more techniques. The authors employed hybrid approach by combining the Rule based approach and Hidden Markov Model in succession. The system classifies entities to three classes namely person, location and organization names respectively. The system can solve ambiguous, nested entities in the sentence. The NER system proposed is quite efficient in identifying named entities and with limited size corpora it achieves high F-measure score[10].

Prakash Hiremath and Shambhavi B. R in their paper made a study on various approaches and methods for named entity recognition in Indian languages[11]. Some of them are mentioned here. One approach is the Support Vector Machine, which is a supervised machine learning algorithm with learning algorithms that analyse data and recognize patterns associated, and is used for the purpose of classification and analysing relationships among the variables. Another approach is using Hidden Markov Model[10], the probabilistic finite state models having parameters for state-transition probabilities and state specific observation probabilities in which the system being modelled is assumed to be a Markov process with unobserved state. Each state has a probability distribution over the possible output entities. CRF is a probabilistic approach for recognising and learning structured data. It can handle inflections in the words and nested tagging of the named entities. In a Hybrid approach, any two approaches are combined together so as to produce more accuracy. The rule-based and machine learning-based approaches are combined and achieve the strongest points from each method.

III. PROPOSED SYSTEM

The system proposed in this paper is Named Entity Recognition for Malayalam language using Fuzzy support vector machine. The named entities in a Malayalam text[12] is identified by contextual semantic features and linguistic grammar features. Support Vector Machine(SVM) is a supervised machine learning approach for classification of statistical data. The classification is challenging when the features results ambiguous outputs. The name of a person and the name of an organization can be same, or the pattern of the naming and the features for the names may be either same or can possess similar formatting. This similarities tends to result in multi-classification of entities by the SVM classifier. To handle the ambiguity, in the proposed approach, a fuzzy module is

implemented along with the SVM classifier, since in normal classification methods each named entity belongs to a fixed class based on its features[7].

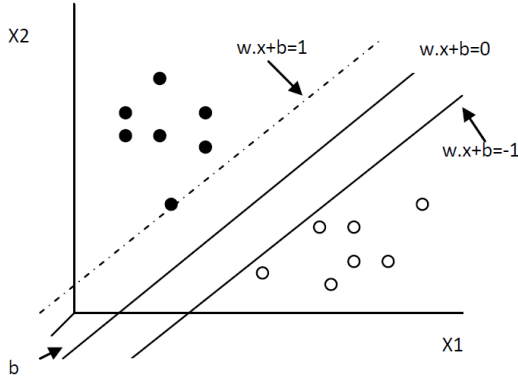


Fig. 1. Classification by Hyperplanes in SVM

A. Support Vector Machine

SVM[6] is one of the famous supervised machine learning algorithms for binary classification in all various data set and it gives the best results where the data set is separable and especially when the training data set is a few. The goal of a supervised SVM classifier is to produce a model which predicts target value of data instances in the testing set, when given only the attributes (features)[7]. Each of the training sample data, is labelled with either positive or negative class tag, as: $(x_1, y_1) \dots (x_n, y_n)$, where $x_i \in \mathbb{R}^n$, $y_i \in \{+1, -1\}$ where x_i is a feature vector of the i th example represented by y and n dimensional vector. y_i is the label of the i th example, (either +1 for positive or -1 for negative). N is the total number of training examples derived from the training set. In basic form, a SVM learns to find a linear hyperplane that separate both positive and negative examples with maximal margin. The maximal margin[7] can be expressed as follows:

$$(w \cdot x) + b = 0, (w \in \mathbb{R}^n, b \in \mathbb{R}) \quad (1)$$

The Fig. 1 depicts the classification in the SVM classifier using the hyperplanes. The hyperplane separates the training data into positive and negative parts, such that: $y_i(w \cdot x_i) \geq 1$. However, several of such separating hyperplane exists and SVM finds the optimal hyperplane that maximizes the margins between the nearest examples to the hyperplane. For the best classification we have to choose the best hyperplane that is the one that represents the largest separation, or margin, between the two classes. The hyperplane created is used for the classification by the SVM classifier. The test vectors are divided into two sets by the classifier using the hyperplane as the vectors on one side of the hyperplane to one class and the vectors on the other side to other class. One of the classes is defined as positive and the other as negative class. Thus the classification of the test data members is done by the SVM classifier using the hyperplane.

The testing and training of the SVM classifier along with the description of the features used in the proposed system are described in the later section.

B. Fuzzy Support Vector Machine

Fuzzy logic[13] is a form of many-valued logic in which the truth values of variables may be any real number between 0 and 1. Fuzzy logic has been extended to handle the concept of partial truth, where the truth value may range between completely true and completely false. Furthermore, when linguistic variables are used, these degrees may be managed by specific functions or rules. A fuzzy rule is written as If situation Then conclusion. The situation, called rule premise or antecedent, is defined as a combination of relations such as x is A for each component of the input vector. The conclusion part is called consequence or conclusion. SVM causes ambiguity in classifying the named entities to predefined classes. To handle this weakness point of multi classification of SVM, fuzzy logic module is integrated with the SVM. When multi-classification occurs the fuzzy module checks for the more accurate class for the ambiguous named entity and assigns the entity to the one which is more accurate.

C. Methodology: Fuzzy SVM based Named Entity Recognition

The main steps involved in the proposed system are: Preprocessing of the text, then training the SVM classifier with the training data set and then testing the classifier with the test data set during which fuzzy logic is applied with the classifier so as to make the system more efficient by handling the ambiguity in the classification which results in multi-classification. The system design is shown in the figure(Fig. 2).

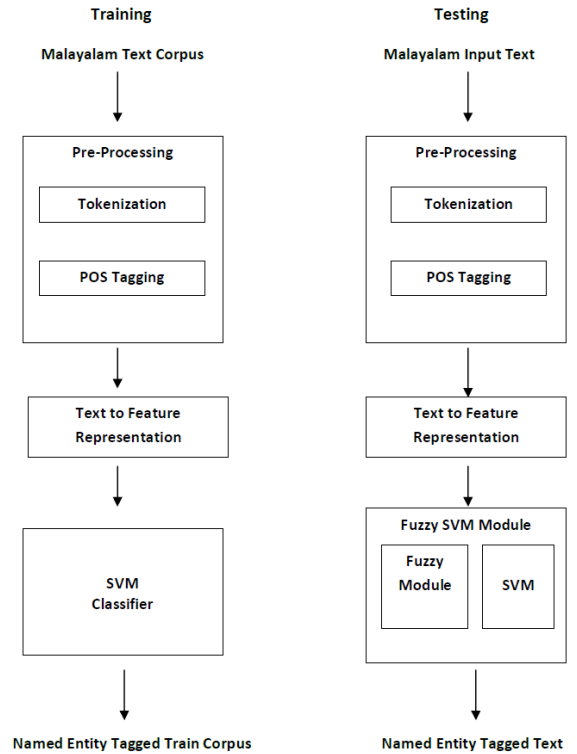


Fig. 2. NER in Malayalam Languages using Fuzzy SVM

The Malayalam input text is first applied for preprocessing. Preprocessing procedure includes tokenization[14] and pos

tagging. POS tagging[15] is done by implementing TnT tagger. After preprocessing, the features are selected for training the SVM classifier. The features of the SVM classifier for the proposed system are:

- 1 POS Tags: The TnT tagger is trained with a specific tag set so that it can act as a feature for the SVM classifier. Named entities are most probably nouns. So the entities with NN tags are extracted. Also TnT tag set has NNO for organization, NNP for places, NNM for months, NNL for words related to location and NQ for the digit informations. POS tag feature is a binary valued feature where the mentioned tags are given 1 if present and 0 if not.
- 2 Word prefix/ suffix: A list of prefixes and suffixes for the named entities is to be constructed. The list may contains the suffixes like -sherry, -ooru, -puram, -puzha, -kode, -kulam, -kaadu etc related to place entity class and -pilla, -varma, -shekhar, -gandhi etc related to person name entity class and -aam, -maasam, -thiyati, -naal, -aazcha etc for related to time entity class.
- 3 Digit information: This is a binary valued feature where the feature value is 1, if the word is numerical and 0 if not.
- 4 Surrounding words: The context word information and their tags are another important feature. For this different window sizes are used like, unigrams, bigrams and trigrams. For example, if the digit information feature has value 1 and if its context word has the tag NNM, it can be classified as the time entity class.

Input Text:

ഇന്ത്യയുടെ അഭിമാനമായ ഐ.എസ്.ആർ.ഓയിൽ ശാസ്ത്രജ്ഞൻ ആയിരുന്ന ശ്രീ അബ്ദുൽകലാം ജനിച്ചത് 1931 ഒക്ടോബർ 15ന് ആണ് .

Preprocessing Step:

Tokenization:

ഇന്ത്യയുടെ അഭിമാനമായ ഐ.എസ്.ആർ.ഓയിൽ ശാസ്ത്രജ്ഞൻ
ആയിരുന്ന ശ്രീ അബ്ദുൽകലാം ജനിച്ചത് 1931 ഒക്ടോബർ 15ന്
ആണ് .

POS Tagging:

ഇന്ത്യയുടെ\NN അഭിമാനമായ\NN ഐ.എസ്.ആർ.ഓയിൽ\NNO
ശാസ്ത്രജ്ഞൻ\NNO ആയിരുന്ന\VM ശ്രീ\NN അബ്ദുൽകലാം\NN
ജനിച്ചത്\NN 1931\WQ ഒക്ടോബർ\NNM 15ന്\WQ
ആണ്\NN .\SYM

NE Tagged Output Text:

ഇന്ത്യയുടെ<Place> അഭിമാനമായ ഐ.എസ്.ആർ.ഓയിൽ<Organization>
ശാസ്ത്രജ്ഞൻ ആയിരുന്ന ശ്രീ അബ്ദുൽകലാം<Name> ജനിച്ചത് 1931
ഒക്ടോബർ 15ന്<Date> ആണ് .

Fig. 3. Example of the working of the system

A set of 1000 Malayalam sentences are selected from the online Malayalam news articles, blogs etc. SVM

classifier is modelled with the above features. The POS tagged input text is converted to feature vectors. The implementation of the SVM classifier has two phases: Training phase and the Testing phase. During the training phase, data features and data labels are used. At the testing phase, SVM classifier takes data features and outputs the class to which the they belongs to. The predefined classes are : Person Name, Place, Organization and Date.

The SVM classifier is trained with the sentences by giving the binary values(0/1) for each of the selected features. The values are given, as if the feature is true then feature value is 1 and otherwise 0. Thus in this way for each input sentence a set of values are obtained, this is called a feature vector. And a set of feature vectors corresponding to the training set is created. For training the SVM classifier, with the feature vector, the data label or the class to which the sentence is to be classified must be represented. This can be done either manually or through the program. Now the SVM classifier is trained with the set of feature vectors and the corresponding data labels.

During the testing phase, the SVM classifier is integrated with the fuzzy module. The integration of fuzzy module with the support vector machine classifier makes the system as a whole capable to handle any multi classification, if occurs. The output of the SVM classifier is analyzed and check if a named entity belongs to more than one class. The ambiguity occurs between the name class and the organization class. This is due to the fact that name of a person and an organization name can be same and they may follow similar formatting. The main feature used is the presence of action verbs which probably denotes the action of the subject in the context. When the verb is action verb, then the subject is probably a human that is a person name. The subject-object-verb of the text can be distinguished by using a parser. By following this approach, the ambiguity is resolved and the entity is classified to the most appropriate class. Finally the NEs are tagged with the most accurate class and the NE tagged output is produced. The Fig. 3 demonstrate the working of the proposed NER system.

IV. EVALUATION RESULTS

The proposed system is firstly trained for a large set of Malayalam sentences. The set contains a set of 1000 malayalam sentences collected from the online Malayalam news papers, blogs etc. Then it is tested for a set of 500 input sentences, which includes the sentences other than those in the train data corpus. The test data is given to the proposed NER model to classify it into predefined classes namely, person name, organization, place and date. The evaluation is done in the system with SVM classifier alone and then the system with SVM classifier integrated with fuzzy module. The two evaluation results are compared. The two evaluation results are tabulated below. The performance of the system is measured by the F-measure value. The F-measure is computed using two parameters: Precision and Recall[10].

Precision: Precision is ratio of the number of tags in

the text marked up by our system correctly to the total number of tags being marked up. It is the percentage of output tags generated in the output text that are correct. It is denoted as P[10].

$$Precision = CorrectNEs/TotalNEs[8] \quad (2)$$

which is identified by the System

Recall: Recall is ratio of the number of tags in the text marked up by our system correctly to the total number of tags present in the text. It is the percentage of correct tags that are marked in the output text. It is denoted as R. The main purpose of recall is to measure how well our system can perform the recognition of entity names[8].

$$Recall = CorrectNEs/TotalstandardNEs[8] \quad (3)$$

which is in the System

F-Measure: F-Measure score is calculated using precision(P) and Recall(R). It is the harmonic mean of the precision and recall. F-score is computed as:

$$Fscore = 2 * P * R / P + R[6] \quad (4)$$

$$Recall = CorrectNEs/TotalstandardNEs[8] \quad (5)$$

which is in the System

TABLE I: Evaluation Table of the System with SVM Classifier

NE Class	Precision(P)	Recall(R)	F-Score(F)
Date	95.34	85.41	90.10
Name	83.33	72.91	77.77
Place	91.95	83.33	87.42
Organization	85.71	75	79.99

In this paper, the performance of the system at its two stages are evaluated and the comparison is done. The evaluation is done in this way so as to compare the performance of the NER system with SVM classifier and the NER system with fuzzy SVM classifier. The precision, recall and F-measure of the proposed system when only SVM classifier is employed and that when the system is integrated with fuzzy module with the SVM classifier are tabulated in the TABLE II.

The performance of the system is compared using the F measure value. From the table2 it is clear that the performance of the system with SVM classifier is 83.82module is integrated with the system, it raises to 88.13This increase in performance is because the fuzzy module could handle the multi-classification between name class and organization class from the SVM classifier. The graph showing the performance is also

TABLE II: Evaluation Table of the System with Fuzzy SVM classifier

Malayalam NER	Precision	Recall	F-Score
With SVM	89.08	79.16	83.82
With Fuzzy-SVM	92.85	83.87	88.13

shown in the Fig. 4. In the bar graph the X-axis represents the NER systems that are compared and the Y-axis represents the precision, recall and F-score values.

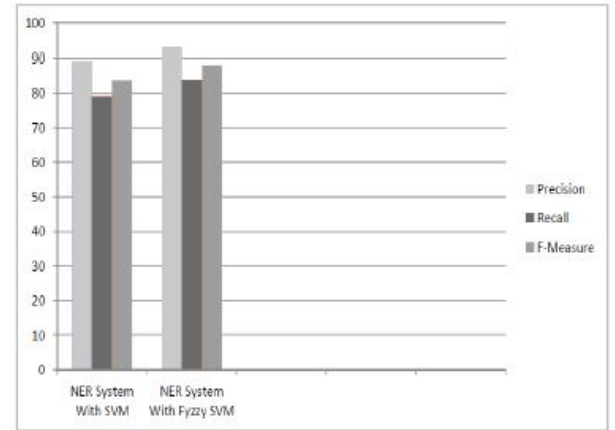


Fig. 4: Performance Evaluation Results

V. CONCLUSION AND FUTURE SCOPE

The major Natural Language Processes(NLP) are Information Extraction, Data Classification, Text Summarization, Text Categorization etc. In all these applications Named Entity Recognition can be a subtask. The system proposed in the paper is an automated named entity recognizer which can recognise an atomic element as named entity and classify it to a predefined class using machine learning approach. The classes defined in the proposed system are person name(Name), name of organization(Organization), name of place(Place) and the date(Date). The system uses support vector machine for the classification of the entities. The SVM uses semantic and contextual features of the input text to identify the classes. Malayalam, being an agglutinative and morphologically rich language produce ambiguities in the results if the SVM classifier by generating multi-classification anomalies. Multi-classification is a specific case of the SVMs output which is the classification of the same entity to more than one class. This situation is controlled and the performance of the system is improved to a great extend by using the fuzzy module. The fuzzy module handles the ambiguities between the name and organization class in the results of the SVM classifier certain fuzzy rules. Fuzzy rules perform the

decision making by identifying the subject in the target sentence by considering the verbs in it, whether it is an action verb or not. From the evaluation results of the system, it is clear that the fuzzy module can increase the performance of the system by resolving the uncertainty in multi-classification of the SVM classifier. So this idea can be used to implement a domain-independent Named Entity Recognition system, which can classify named entities of a Malayalam text to a set of NE classes, that can handle the ambiguity between any two classes of the system. Future developments include improving the entity identification overhead in the molecular biology, bioinformatics etc. Also, in reducing the annotation labour by employing semi-supervised learning.

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